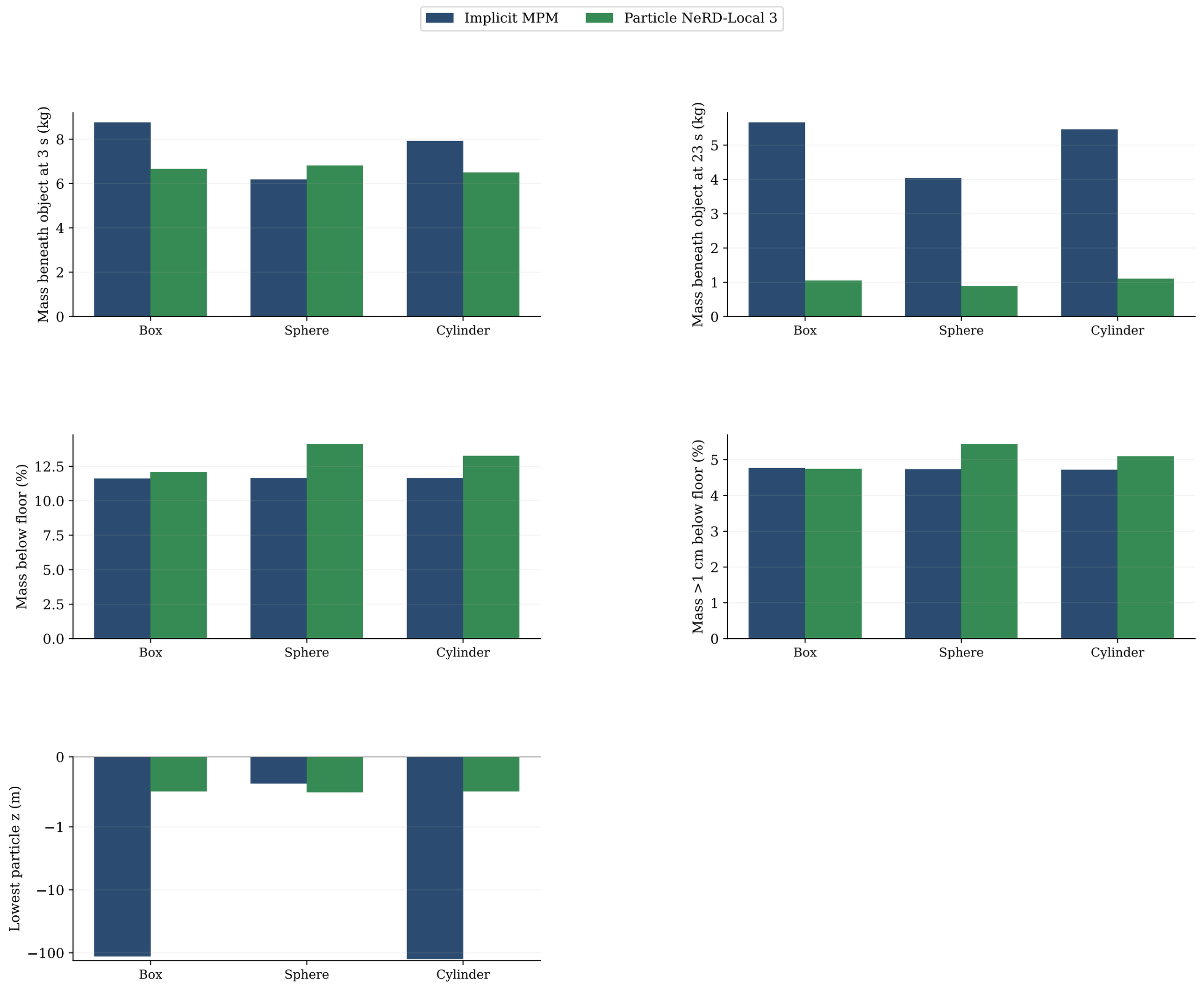

Figure 8: Frozen object-model checkpoint 4,500. Open circles mark measured zeros. Top row: physical mass beneath the object at 3 and 23 s. The region is its world x/y bounding rectangle extending 0.10 m below the lowest body point, clipped above the floor. Middle row: mass with centers below the z=−0.30 m floor (1 µm tolerance), and mass more than 1 cm below it. Last panel: lowest particle-center z at 23 s, with symmetric-log scale (linear within ±1 m). Percentages use total represented physical mass. One paired run per shape; all escaped particles are retained. MPM also settles and leaks.